Ecuaciones en derivadas parciales.

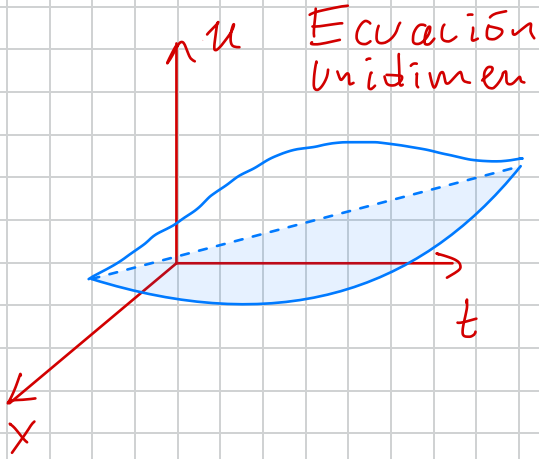
Sea u una variable dependiente de $\mathbf{X} = (x_1, x_2, \dots, x_n)$ y t.

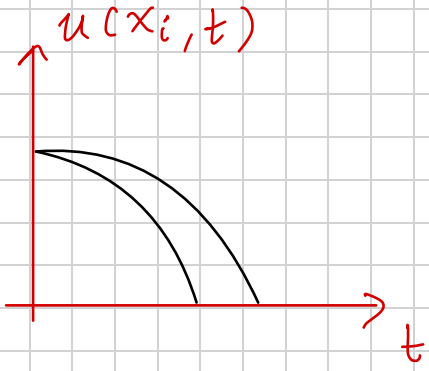
Posición — tiempo.

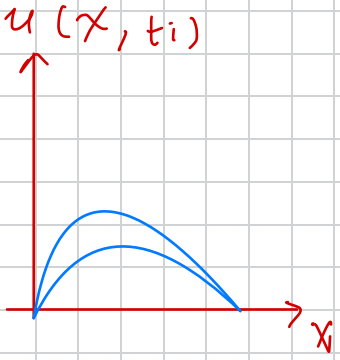
$$K \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} \; ; \quad K > 0 \quad \text{Parabólica.}$$

Ecuación de calor unidimensional

$u(x,t)$







$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

Hiperbólica.

Ecuación de onda unidimensional.

Elíptica

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Laplaciano en dos dimensiones

$$; \quad \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

$$\nabla u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right)$$

$$\nabla \cdot \nabla u = \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

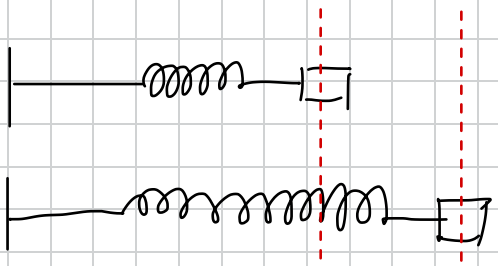
Laplaciano en tres dimensiones.

Definición: condición inicial.

Para un valor de $t=0$ se espresa

$u(x,0) = f(x)$ o $\left.\frac{\partial u}{\partial t}\right|_{t=0} = g(x)$. Donde $f(x)$ representa la distribución de u para el instante de tiempo $t=0$. Esta distribución satisface la EDP en el instante inicial $u(x,0) = f(x)$ para $x \in [0, L]$.

$\left.\frac{\partial u}{\partial t}\right|_{t=0} = g(x)$ Velocidad inicial.



Δx

$$W = \frac{1}{2} k \Delta x^2$$

$$W = \frac{1}{2} k x^2$$

$$\frac{\partial W}{\partial x} = \underbrace{kx}_{\text{Fuerza del resorte.}}$$

Definición condición de frontera:

Alguno de los extremos $x \in [0, L]$ tendrá que estar fijo o ambos

$$u(0,t) = 0 \quad ; \quad u(L,t) = 0 \quad , t > 0$$

Resolver la EDP: Ecuación diferencial de calor unidimensional.

$$K\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}\ ; \qquad K>0$$

Sujeta a: Condiciones iniciales

$$u(x,0) = f(x)\ ; \quad 0<x<L$$

Condiciones de frontera

$$u(0,t) = f(x) \qquad t>0$$

$$u(L,t) = g(x) \qquad t>0$$

Resolver la EDP: Ecuación diferencial de onda unidimensional.

$$a^2\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}\ ;$$

Sujeta a: Condiciones iniciales

$$u(x,0) = f(x)\ ; \quad 0<x<L$$

$$\left.\frac{\partial u}{\partial t}\right|_{t=0} = g(x)\ ; \quad 0<x<L.$$

Condiciones de frontera

$$u(0,t) = f(x) \qquad t>0$$

$$u(L,t) = g(x) \qquad t>0$$

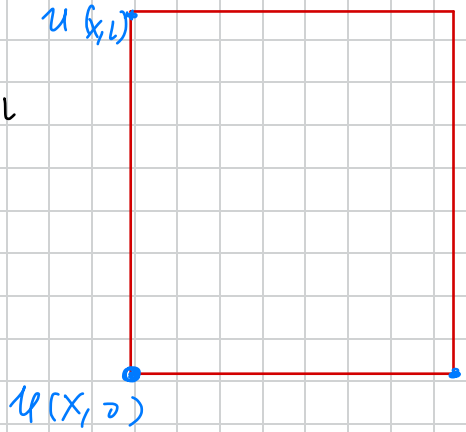
Resolver la EDP: Laplaciana en dos dimensiones

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0\ ;$$

Sujeta a: Condiciones de frontera

$$\left.\frac{\partial u}{\partial x}\right|_{x=0} \qquad \left.\frac{\partial u}{\partial x}\right|_{x=L}$$

$$u(x,0) \qquad u(x,L)$$



Variación en la ecuación de onda.

$$a^2 \frac{\partial^2 u}{\partial x^2} + f(x,t) - c\frac{\partial u}{\partial t} - ku = \frac{\partial^2 u}{\partial t^2}$$

Fuerza de restablecimiento

Fuerza externa

Amortiguamiento

Solución EDP de calor:

EDP: $k\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$; $0 < x < L$; $t > 0$.

C.F: $u(0,t) = 0$; $u(L,t) = 0$; $t > 0$

C.I: $u(x,0) = f(x)$; $0 < x < L$.

$$u(x,t) = X(x)\,T(t)$$

$$\frac{\partial u}{\partial t} = X(x)\,T'(t)$$

$$\frac{\partial^2 u}{\partial x^2} = X''(x)\,T(t)$$

$$k\,X''(x)\,T(t) = X(x)\,T'(t)$$

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{kT(t)} = -\lambda$$

En la frontera $u(0,t) = X(0)\,T(t) = 0$

$u(L,t) = X(L)\,T(t) = 0$

Para que sea valido $X(0) = X(L) = 0$

$$\frac{X''(x)}{X(x)} = -\lambda$$

$$X''(x) = -\lambda X(x)$$

$$X''(x) + \lambda X(x) = 0$$

De acuerdo con la clase 24 sept 2026 pg 2, 3, 4.

La Solución de $X''(x) + \lambda X(x) = 0$

$X(x) = c_1 + c_2 x \quad ; \quad \lambda = 0$ (*)

$X(x) = c_1 e^{-2\alpha x} + c_2 e^{2\alpha x} \quad ; \quad \lambda = -\alpha^2 < 0$ (*) (**)

$X(x) = c_1 \cos(\alpha x) + c_2 \operatorname{Sen}(\alpha x) \quad ; \quad \lambda = \alpha^2 > 0$ (*) (**) (***)

En (*)

$X(0) = c_1 + c_2(0) = 0 \quad ; \quad c_1 = 0$

$X(L) = c_1 + c_2 L = 0 \quad ; \quad c_2 = 0$

$u(x,t) = 0$ Solución trivial.

En (*) (**)

$X(0) = c_1 + c_2 = 0$

$X(L) = c_1 e^{-2\alpha L} + c_2 e^{2\alpha L} = 0$

$$\begin{bmatrix} 1 & 1 \\ e^{-2\alpha L} & e^{2\alpha L} \end{bmatrix} \begin{vmatrix} c_1 \\ c_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

$c_1 = c_2 = 0$

$u(x,t) = 0$ Solución trivial.

En (*) (**) (***)

$X(0) = c_1 = 0$

$X(L) = c_1 \cos(\alpha L) + c_2 \operatorname{Sen}(\alpha L) = 0$

$c_2 \neq 0$ entonces $\operatorname{Sen}(\alpha L) = 0$

$\alpha L = 0 \quad ; \quad \alpha L = n\pi \quad ; \quad \alpha = \frac{n\pi}{L}$

$X(x) = c_2 \operatorname{Sen}\left(\frac{n\pi}{L} x\right) \quad ;$

$$\frac{T'(t)}{kT(t)} = -\lambda \quad ; \quad T'(t) = -\lambda k T(x)$$

$$T' + \lambda k T = 0$$

$$u = e^{\lambda k \int dt} = e^{\lambda k t}$$

$$\int D\left[e^{\lambda k t} T\right] = \int 0$$

$$e^{\lambda k t} T = c$$

$$T(t) = c e^{-\lambda k t}$$

$$u(x,t) = X(x) T(t) = c_2 \operatorname{Sen}\left(\frac{n\pi}{L} x\right) c e^{-\lambda k t}$$

$$u(x,t) = c c_2 e^{-\lambda k t} \operatorname{Sen}\left(\frac{n\pi x}{L}\right)$$

De la condición $\lambda = \alpha^2$

$$\lambda = \alpha^2 = \left(\frac{n\pi}{L}\right)^2$$

$$\lambda = \frac{n^2\pi^2}{L^2}$$

$$u(x,t) = c c_2 e^{-k\left(\frac{n^2\pi^2}{L^2}\right)t} \operatorname{Sen}\left(\frac{n\pi}{L} x\right)$$

Si $t = 0$, $u(x,0) = f(x)$

$$u(x,0) = c c_2 (1) \operatorname{Sen}\left(\frac{n\pi}{L} x\right) = f(x)$$

$$a_n = c c_2$$

$$u_n(x,0) = f(x) = a_n \operatorname{Sen}\left(\frac{n\pi x}{L}\right)$$

$$u(x,t) = \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} a_n e^{-k(n^2\pi^2/L^2)t} \operatorname{Sen}\left(\frac{n\pi x}{L}\right)$$

El conjunto $\left\{ \operatorname{sen}\left(\frac{n\pi x}{L}\right) \right\}_{n=1}^{\infty}$ es ortogonal

$$(\phi_i, \phi_j) = \int_a^b \phi_i \phi_j = \begin{cases} = 0 & i \neq j \\ \int \phi_i^2 & i = i \end{cases}$$

$$\int_0^L f(x) \operatorname{sen}\left(\frac{n\pi x}{L}\right) = a_n \int_0^L \operatorname{sen}\left(\frac{n\pi x}{L}\right) \cdot \operatorname{sen}\left(\frac{n\pi x}{L}\right)$$

$$a_n = \frac{\int_0^L f(x) \operatorname{sen}\left(\frac{n\pi x}{L}\right) dx}{\int_0^L \operatorname{sen}^2\left(\frac{n\pi x}{L}\right) dx} = \frac{2}{L} \int_0^L f(x) \operatorname{sen}\left(\frac{n\pi x}{L}\right) dx$$

$\frac{1}{2}\left(1 - \cos\left(2\frac{n\pi x}{L}\right)\right)$

$\frac{1}{2}\left(x - \frac{\operatorname{sen}\left(2\frac{n\pi x}{L}\right)}{\frac{2n\pi}{L}}\right)\Big|_0^L$

$\frac{L}{2}$

$$u(x,t) = \frac{2}{L} \sum_{n=1}^{\infty} \left(\int_0^L f(x) \operatorname{sen}\left(\frac{n\pi}{L} x\right) dx \right) e^{-k\left(\frac{n\pi}{L}\right)^2 t} \operatorname{sen}\left(\frac{n\pi}{L} x\right)$$

Si $u(x,0) = 1$ entonces $f(x) = 1$

$$u(x,t) = \frac{2}{L} \sum_{n=1}^{\infty} \left(\int_0^L (1) \operatorname{sen}\left(\frac{n\pi}{L} x\right) dx \right) e^{-k\left(\frac{n\pi}{L}\right)^2 t} \operatorname{sen}\left(\frac{n\pi}{L} x\right)$$

$$= \frac{2}{L} \sum_{n=1}^{\infty} \left(\frac{-L \cos\left(\frac{n\pi}{L} x\right)}{n\pi} \Big|_0^L \right) e^{-k\left(\frac{n\pi}{L}\right)^2 t} \operatorname{sen}\left(\frac{n\pi}{L} x\right)$$

$$\frac{-L}{n\pi}\left(\cos(n\pi) - 1\right)$$

$$\frac{L}{n\pi}\left(1 - (-1)^n\right)$$

$$u(x,t) = \frac{2}{L} \sum_{n=1}^{\infty} \left(\frac{L}{n\pi}\left(1 - (-1)^n\right) \right) e^{-k\left(\frac{n\pi}{L}\right)^2 t} \operatorname{sen}\left(\frac{n\pi}{L} x\right)$$

Solución de la ecuación de Onda.

$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad ; \quad 0 < x < L \; ; \; t > 0$$

$$u(0,t) = 0 \; ; \quad u(L,t) = 0 \quad ; \quad t > 0$$

$$u(x,0) = f(x) \; ; \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = g(x) \; ; \quad 0 < x < L$$

$$u(x,t) = X(x)\,T(t) \qquad a^2 X''(x) T(t) = X(x) T''(t)$$

$$\frac{\partial^2 u}{\partial t^2} = X(x)\,T''(t) \qquad \frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

$$\frac{\partial^2 u}{\partial x^2} = X''(x)\,T(t)$$

$$X'' + \lambda X(x) = 0$$

$$X(x) = c_2 \operatorname{Sen}\left(\frac{n\pi}{L} x\right) \; ; \qquad D^2 + a^2\lambda = 0$$

$$D^2 = -a^2\lambda$$

$$T'' + a^2 \lambda T = 0 \qquad D = 0 \pm j a\sqrt{\lambda}$$

$$\lambda = \alpha^2 > 0 \quad (\alpha)(x)(t)$$

$$T = c_1 \cos(a\sqrt{\lambda}\, t) + c_2 \operatorname{Sen}(a\sqrt{\lambda}\, t)$$

$$\sqrt{\lambda} = \sqrt{\alpha^2} = \sqrt{\left(\frac{n\pi}{L}\right)^2}$$

$$T(t) = c_1 \cos\left(a\frac{n\pi}{L} t\right) + c_3 \operatorname{Sen}\left(a\frac{n\pi}{L} t\right)$$

$$u_n = c_2 \operatorname{Sen}\left(\frac{n\pi}{L} x\right)\left(c_1 \cos\left(a\frac{n\pi}{L} t\right) + c_3 \operatorname{Sen}\left(a\frac{n\pi}{L} t\right)\right)$$

$$u(x,t) = \sum_{n=1}^{\infty} c_2 \operatorname{Sen}\left(\frac{n\pi}{L} x\right)\left(c_1 \cos\left(a\frac{n\pi}{L} t\right) + c_3 \operatorname{Sen}\left(a\frac{n\pi}{L} t\right)\right)$$

$$u(x,0) = \sum_{n=1}^{\infty} \underbrace{c_1 c_2}_{A_n} \operatorname{Sen}\left(\frac{n\pi}{L} x\right) = f(x)$$

$$u(x,0) = \sum_{n=1}^{\infty} \underbrace{C_1 C_2}_{A_n} \operatorname{Sen}\left(\frac{n\pi}{L}x\right) = f(x)$$

$$\int_0^L f(x)\,\operatorname{Sen}\left(\frac{n\pi x}{L}\right) dx = \int_0^L \sum_{n=1}^{\infty} A_n \operatorname{Sen}\left(\frac{n\pi x}{L}\right).\operatorname{Sen}\left(\frac{n\pi x}{L}\right) dx$$

$$A_n = \frac{2}{L}\int_0^L f(x)\,\operatorname{Sen}\left(\frac{n\pi x}{L}\right) dx$$

$$B_n = C_1 C_3$$

$$u(x,t) = \sum_{n=1}^{\infty} \operatorname{Sen}\left(\frac{n\pi}{L}x\right)\left(A_n \cos\left(a\frac{n\pi}{L}t\right) + B_n \operatorname{Sen}\left(a\frac{n\pi}{L}t\right)\right)$$

$$\frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} \operatorname{Sen}\left(\frac{n\pi x}{L}\right)\left(-A_n \operatorname{Sen}\left(\frac{an\pi}{L}t\right)\frac{dn\pi}{L} + B_n \cos\left(\frac{an\pi}{L}t\right)\frac{an\pi}{L}\right)$$

$$\left.\frac{\partial u}{\partial t}\right|_{t=0} = \sum_{n=1}^{\infty} \operatorname{Sen}\left(\frac{n\pi x}{L}\right)\left(B_n \frac{an\pi}{L}\right) = g(x)$$

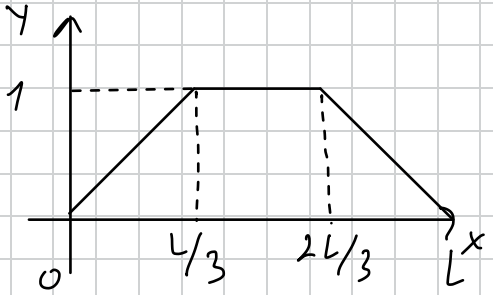
$* \operatorname{Sen}\left(\frac{n\pi x}{L}\right)$

$$\int_0^L g(x)\operatorname{Sen}\left(\frac{n\pi x}{L}\right) dx = B_n \frac{an\pi}{L}\underbrace{\int_0^L \operatorname{Sen}^2(n\pi x/L)\,dx}_{L/2}$$

$$B_n = \frac{2}{n\pi a}\int_0^L g(x)\operatorname{Sen}\left(\frac{n\pi x}{L}\right) dx$$

Tarea: Calcular A_n y B_n Si

$$f(x) =$$



$$g(x) = \operatorname{Sen}(2x).$$